Chapter 8. Canonical Correlation Analysis

Maximal correlation between two sets of variables

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1 Setting

We want to summarize the association between two sets of variables $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$ with $\operatorname{Cov}\begin{pmatrix}\mathbf{X}\\ \mathbf{Y}\end{pmatrix} = \begin{pmatrix}\mathbf{\Sigma}_{11} & \mathbf{\Sigma}_{12}\\ \mathbf{\Sigma}_{21} & \mathbf{\Sigma}_{22}\end{pmatrix}$, $\mathbf{\Sigma}_{11}, \mathbf{\Sigma}_{22} \succ 0$. Instead of inspecting pq correlations, maximize the correlation of linear combinations

$$\xi = \mathbf{g}^\top \mathbf{X}, \qquad \omega = \mathbf{h}^\top \mathbf{Y}.$$

For $q = 1$ this is the multiple correlation (Ch. 3). PCA maximizes variance *within* one set; CCA maximizes correlation *between* two sets.

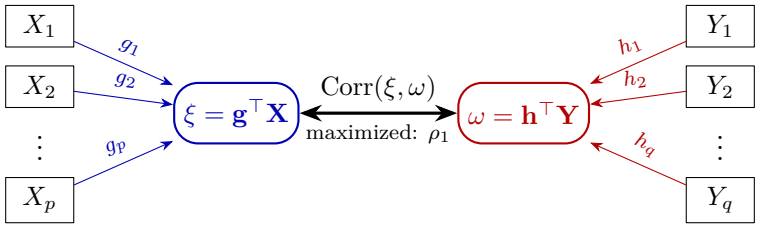


Figure. *CCA projects each set onto one direction, $\mathbf{X} \mapsto \xi = \mathbf{g}^\top \mathbf{X}$ and $\mathbf{Y} \mapsto \omega = \mathbf{h}^\top \mathbf{Y}$, and chooses $(\mathbf{g}, \mathbf{h})$ so that the two projections are as correlated as possible. Later pairs repeat this subject to being uncorrelated with the earlier variates of their own set.*

Definition 8.1: canonical variates and canonical correlations

$$(\mathbf{g}_1, \mathbf{h}_1) = \arg \max_{\mathbf{g} \in \mathbb{R}^p,\, \mathbf{h} \in \mathbb{R}^q} \mathrm{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}),$$

$$(\mathbf{g}_k, \mathbf{h}_k) = \arg \max_{\substack{\mathbf{g} \in \mathbb{R}^p,\, \mathbf{h} \in \mathbb{R}^q \\ \mathbf{g}^\top \mathbf{\Sigma}_{11} \mathbf{g}_j = 0,\ \mathbf{h}^\top \mathbf{\Sigma}_{22} \mathbf{h}_j = 0,\ j = 1, \ldots, k-1}} \mathrm{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}).$$

$\xi_k = \mathbf{g}_k^\top \mathbf{X}$, $\omega_k = \mathbf{h}_k^\top \mathbf{Y}$ form the kth pair of canonical variates and $\rho_k = \mathrm{Corr}(\xi_k, \omega_k)$ is the kth canonical correlation. Correlation is scale-free, so normalize $\mathrm{Var}(\xi_k) = \mathrm{Var}(\omega_k) = 1$. The constraints mean $\mathrm{Cov}(\xi_k, \xi_j) = \mathrm{Cov}(\omega_k, \omega_j) = 0$, and $\rho_k \le \rho_j$ for $j < k$.

$\rho(\mathbf{X}, \mathbf{Y}) := \max \mathrm{Corr}(\mathbf{a}^\top \mathbf{X}, \mathbf{b}^\top \mathbf{Y}) = \rho_1$ is the **largest canonical coefficient**.

2 Solution via the Singular Value Decomposition

Motivation. After each set is whitened to identity covariance, correlation becomes a bilinear form $\mathbf{a}^\top \mathbf{M} \mathbf{b}$ in two unit vectors, and maximizing a bilinear form over unit vectors is exactly what the singular value decomposition does.

Theorem 8.2: solution of CCA

Let $\mathbf{M} = \mathbf{\Sigma}_{11}^{-1/2} \mathbf{\Sigma}_{12} \mathbf{\Sigma}_{22}^{-1/2}$ $(p \times q)$ have SVD $\mathbf{M} = \sum_{k=1}^{r} d_k \mathbf{a}_k \mathbf{b}_k^\top$, $d_1 \ge \cdots \ge d_r > 0$, $r = \operatorname{rank} \mathbf{\Sigma}_{12}$. Then

$$\rho_k = d_k, \qquad \mathbf{g}_k = \mathbf{\Sigma}_{11}^{-1/2} \mathbf{a}_k, \qquad \mathbf{h}_k = \mathbf{\Sigma}_{22}^{-1/2} \mathbf{b}_k.$$

Equivalently $\mathbf{a}_k$ is the kth eigenvector of $\mathbf{M}\mathbf{M}^\top = \mathbf{\Sigma}_{11}^{-1/2} \mathbf{\Sigma}_{12} \mathbf{\Sigma}_{22}^{-1} \mathbf{\Sigma}_{21} \mathbf{\Sigma}_{11}^{-1/2}$, $\mathbf{b}_k$ that of $\mathbf{M}^\top \mathbf{M} = \mathbf{\Sigma}_{22}^{-1/2} \mathbf{\Sigma}_{21} \mathbf{\Sigma}_{11}^{-1} \mathbf{\Sigma}_{12} \mathbf{\Sigma}_{22}^{-1/2}$, and ρ_k^2 is the eigenvalue.

Proof. Substituting $\mathbf{a} = \mathbf{\Sigma}_{11}^{1/2} \mathbf{g}$, $\mathbf{b} = \mathbf{\Sigma}_{22}^{1/2} \mathbf{h}$,

$$\mathrm{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}) = \frac{\mathbf{g}^\top \mathbf{\Sigma}_{12} \mathbf{h}}{\sqrt{\mathbf{g}^\top \mathbf{\Sigma}_{11} \mathbf{g}} \sqrt{\mathbf{h}^\top \mathbf{\Sigma}_{22} \mathbf{h}}} = \frac{\mathbf{a}^\top \mathbf{M} \mathbf{b}}{\sqrt{\mathbf{a}^\top \mathbf{a}} \sqrt{\mathbf{b}^\top \mathbf{b}}},$$

so we maximize $\mathbf{a}^\top \mathbf{M}\mathbf{b}$ over $\|\mathbf{a}\| = \|\mathbf{b}\| = 1$. With $\mathbf{M} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$, $\boldsymbol{\alpha} = \mathbf{U}^\top\mathbf{a}$, $\boldsymbol{\beta} = \mathbf{V}^\top\mathbf{b}$, Cauchy–Schwarz gives

$$\mathbf{a}^\top \mathbf{M}\mathbf{b} = \mathbf{a}^\top \mathbf{U}\mathbf{D}\mathbf{V}^\top \mathbf{b} = \sum_k d_k \alpha_k \beta_k \le d_1 \sum_k |\alpha_k \beta_k| \le d_1 \|\boldsymbol{\alpha}\| \|\boldsymbol{\beta}\| \le d_1,$$

with equality at $\mathbf{a} = \mathbf{a}_1$, $\mathbf{b} = \mathbf{b}_1$. Under $\mathbf{g}^\top \boldsymbol{\Sigma}_{11} \mathbf{g}_j = \mathbf{a}^\top \mathbf{a}_j = 0$ and $\mathbf{b}^\top \mathbf{b}_j = 0$ $(j < k)$ we have $\alpha_j = \beta_j = 0$ for $j < k$, and the same argument gives maximum d_k at $(\mathbf{a}_k, \mathbf{b}_k)$. ■

Theorem 8.3: properties of canonical variates

1. $\mathrm{Var}(\xi_k) = \mathrm{Var}(\omega_k) = 1$ and $\mathrm{Corr}(\xi_k, \xi_l) = \mathrm{Corr}(\omega_k, \omega_l) = 0$ for $k \neq l$.
2. $\mathrm{Corr}(\xi_k, \omega_l) = 0$ for $k \neq l$, $\mathrm{Corr}(\xi_k, \omega_k) = \rho_k$.
3. Invariance: canonical correlations are unchanged by $\mathbf{X} \mapsto \mathbf{D}_1\mathbf{X} + \mathbf{c}_1$, $\mathbf{Y} \mapsto \mathbf{D}_2\mathbf{Y} + \mathbf{c}_2$ with $\mathbf{D}_i$ non-singular; covariance and correlation matrices give the same ρ_k.
4. For $q = 1$, $\rho_1^2 = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}/\sigma_{22}$ is the squared multiple correlation.

Proof. (1) $\mathrm{Cov}(\xi_k, \xi_l) = \mathbf{g}_k^\top \boldsymbol{\Sigma}_{11} \mathbf{g}_l = \mathbf{a}_k^\top \mathbf{a}_l = \delta_{kl}$.

(2) $\mathrm{Cov}(\xi_k, \omega_l) = \mathbf{g}_k^\top \boldsymbol{\Sigma}_{12} \mathbf{h}_l = \mathbf{a}_k^\top \mathbf{M}\mathbf{b}_l = d_l \mathbf{a}_k^\top \mathbf{a}_l = d_l \delta_{kl}$.

(3) $\mathrm{Corr}(\mathbf{g}^\top \mathbf{D}_1\mathbf{X}, \mathbf{h}^\top \mathbf{D}_2\mathbf{Y}) = \mathrm{Corr}((\mathbf{D}_1^\top \mathbf{g})^\top \mathbf{X}, (\mathbf{D}_2^\top \mathbf{h})^\top \mathbf{Y})$ and $\mathbf{g} \mapsto \mathbf{D}_1^\top \mathbf{g}$ is a bijection, so the maxima agree.

(4) For $q = 1$ the only nonzero eigenvalue of $\mathbf{M}\mathbf{M}^\top$ is $\mathbf{M}^\top\mathbf{M} = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}/\sigma_{22}$. ■

Theorem 8.4: interpreting the canonical variates

1. With $\mathbf{G} = [\mathbf{g}_1, \ldots, \mathbf{g}_r]$, $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_r]$, $\boldsymbol{\xi} = \mathbf{G}^\top\mathbf{X}$ and $\boldsymbol{\omega} = \mathbf{H}^\top\mathbf{Y}$,

$$\begin{aligned} \mathrm{Cov}(\mathbf{X}, \boldsymbol{\xi}) &= \boldsymbol{\Sigma}_{11}\mathbf{G}, & \mathrm{Cov}(\mathbf{Y}, \boldsymbol{\omega}) &= \boldsymbol{\Sigma}_{22}\mathbf{H}, \\ \mathrm{Cov}(\mathbf{X}, \boldsymbol{\omega}) &= \boldsymbol{\Sigma}_{12}\mathbf{H}, & \mathrm{Cov}(\mathbf{Y}, \boldsymbol{\xi}) &= \boldsymbol{\Sigma}_{21}\mathbf{G}, \end{aligned}$$

   and the correlations follow by dividing row i by $\sqrt{\sigma_{ii}}$ (the variates have unit variance). For standardized variables these are the **canonical loadings** used to name the variates.
2. $\rho_1 \ge \max_{i,j} |\mathrm{Corr}(X_i, Y_j)|$, and more generally $\rho_1 \ge$ the multiple correlation of any single X_i with $\mathbf{Y}$ (and of any Y_j with $\mathbf{X}$).

Proof. (1) $\mathrm{Cov}(\mathbf{X}, \mathbf{G}^\top\mathbf{X}) = \mathrm{Cov}(\mathbf{X})\mathbf{G} = \boldsymbol{\Sigma}_{11}\mathbf{G}$ and $\mathrm{Cov}(\mathbf{X}, \mathbf{H}^\top\mathbf{Y}) = \mathrm{Cov}(\mathbf{X}, \mathbf{Y})\mathbf{H} = \boldsymbol{\Sigma}_{12}\mathbf{H}$; the other two are the same computation.

(2) ρ_1 is the maximum of $\mathrm{Corr}(\mathbf{g}^\top\mathbf{X}, \mathbf{h}^\top\mathbf{Y})$ over all $(\mathbf{g}, \mathbf{h})$. Taking $\mathbf{g} = \mathbf{e}_i$ and maximizing over $\mathbf{h}$ only gives the multiple correlation of X_i with $\mathbf{Y}$, a value attained in a smaller set; taking also $\mathbf{h} = \pm\mathbf{e}_j$ gives $|\mathrm{Corr}(X_i, Y_j)|$. ■

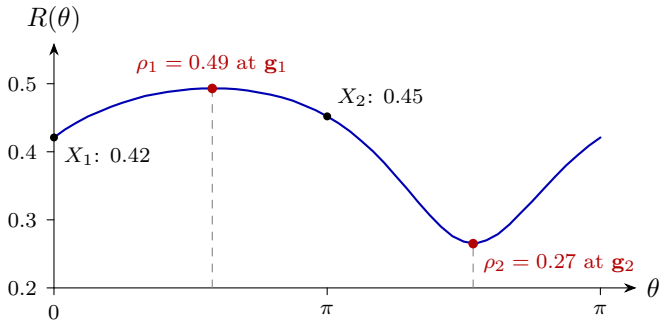


Figure. *A synthetic population with $p = q = 2$ (unit variances, within-set correlations 0.4 in $\mathbf{\Sigma}_{11}$ and 0.2 in $\mathbf{\Sigma}_{22}$, $\mathbf{\Sigma}_{12} = \left(\begin{smallmatrix} 0.30 & 0.35 \\ 0.05 & 0.45 \end{smallmatrix}\right)$). For $\mathbf{g}(\theta) = (\cos\theta, \sin\theta)^\top$, $R(\theta) = \max_{\mathbf{h}} \operatorname{Corr}(\mathbf{g}(\theta)^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y})$ is the multiple correlation of $\mathbf{g}(\theta)^\top \mathbf{X}$ with $\mathbf{Y}$. Its maximum is ρ_1, attained in the direction of $\mathbf{g}_1$, and for $p = 2$ its minimum is ρ_2, attained at $\mathbf{g}_2$. The single variables X_1 ($\theta = 0$) and X_2 ($\theta = \pi/2$) reach only 0.42 and 0.45, as Theorem 8.4(2) requires.*

3 Sample Canonical Correlations

Replace the covariances by $\mathbf{S}_{11} = \widehat{\operatorname{Cov}}(\mathbf{X})$, $\mathbf{S}_{22} = \widehat{\operatorname{Cov}}(\mathbf{Y})$, $\mathbf{S}_{12} = \widehat{\operatorname{Cov}}(\mathbf{X}, \mathbf{Y})$:

$$(\hat{\mathbf{g}}_k, \hat{\mathbf{h}}_k) = \arg \max_{\substack{\mathbf{g} \in \mathbb{R}^p,\, \mathbf{h} \in \mathbb{R}^q \\ \mathbf{g}^\top \mathbf{S}_{11} \hat{\mathbf{g}}_j = 0,\ \mathbf{h}^\top \mathbf{S}_{22} \hat{\mathbf{h}}_j = 0,\ j = 1, \dots, k-1}} \frac{\mathbf{g}^\top \mathbf{S}_{12} \mathbf{h}}{\sqrt{\mathbf{g}^\top \mathbf{S}_{11} \mathbf{g}\, \mathbf{h}^\top \mathbf{S}_{22} \mathbf{h}}}.$$

With $\mathbf{a} = \mathbf{S}_{11}^{1/2} \mathbf{g}$, $\mathbf{b} = \mathbf{S}_{22}^{1/2} \mathbf{h}$ this is maximizing $\dfrac{\mathbf{a}^\top \mathbf{S}_{11}^{-1/2} \mathbf{S}_{12} \mathbf{S}_{22}^{-1/2} \mathbf{b}}{\sqrt{\mathbf{a}^\top \mathbf{a}\, \mathbf{b}^\top \mathbf{b}}} = \max_{\|\mathbf{a}\| = \|\mathbf{b}\| = 1} \mathbf{a}^\top \mathbf{M} \mathbf{b} = \mathbf{a}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top \mathbf{b}$, so

$$\mathbf{a}_i = \mathbf{v}_i(\mathbf{S}_{11}^{-1/2} \mathbf{S}_{12} \mathbf{S}_{22}^{-1} \mathbf{S}_{21} \mathbf{S}_{11}^{-1/2}), \qquad \mathbf{b}_i = \mathbf{v}_i(\mathbf{S}_{22}^{-1/2} \mathbf{S}_{21} \mathbf{S}_{11}^{-1} \mathbf{S}_{12} \mathbf{S}_{22}^{-1/2}),$$

where $\mathbf{v}_i(\mathbf{M})$ is the ith eigenvector of $\mathbf{M}$.

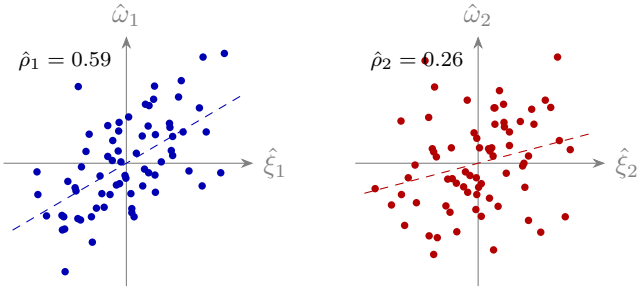


Figure. *Sample canonical variates for $n = 70$ draws from the population of the previous figure ($p = q = 2$). Each variate has sample variance 1, so the dashed least-squares line has slope $\hat{\rho}_k$. The first pair is clearly related; the second, uncorrelated with the first pair, carries the remaining weaker association.*

3.1 Regularized CCA

If p or q is large relative to n, $\mathbf{S}_{11}, \mathbf{S}_{22}$ are singular or unstable and sample canonical correlations are biased towards 1. **Regularized CCA** adds a ridge term,

$$\mathbf{S}_{11} \mapsto \mathbf{S}_{11} + \lambda_1 \mathbf{I}, \qquad \mathbf{S}_{22} \mapsto \mathbf{S}_{22} + \lambda_2 \mathbf{I},$$

with λ_1, λ_2 chosen by cross-validation.

4 Testing Canonical Correlations

Under multivariate normality the likelihood ratio statistic for $H_0 : \mathbf{\Sigma}_{12} = \mathbf{0}$ (all $\rho_k = 0$) is

$$\Lambda^{2/n} = \frac{|\mathbf{S}|}{|\mathbf{S}_{11}||\mathbf{S}_{22}|} = \prod_{k=1}^{\min(p,q)} (1 - \hat{\rho}_k^2),$$

and Bartlett's approximation gives $-(n-1-\frac{p+q+1}{2}) \sum_k \ln(1-\hat{\rho}_k^2) \dot{\sim} \chi^2_{pq}$. The hypothesis that only the first k are nonzero, $H_0^{(k)} : \rho_{k+1} = \cdots = 0$, is tested sequentially with

$$-\Big(n - 1 - \frac{p+q+1}{2}\Big) \sum_{i=k+1}^{\min(p,q)} \ln(1-\hat{\rho}_i^2) \;\dot{\sim}\; \chi^2_{(p-k)(q-k)}.$$

Derivation: the determinant identity

[Supplement] By the block determinant formula, $|\mathbf{S}| = |\mathbf{S}_{22}||\mathbf{S}_{11} - \mathbf{S}_{12}\mathbf{S}_{22}^{-1}\mathbf{S}_{21}| = |\mathbf{S}_{22}||\mathbf{S}_{11}|\,|\mathbf{I} - \hat{\mathbf{M}}\hat{\mathbf{M}}^\top|$, and the eigenvalues of $\hat{\mathbf{M}}\hat{\mathbf{M}}^\top$ are $\hat{\rho}_k^2$, so $|\mathbf{S}|/(|\mathbf{S}_{11}||\mathbf{S}_{22}|) = \prod(1-\hat{\rho}_k^2)$.

5 Relations to PCA, Multiple Correlation and Discriminant Analysis

- **Multiple correlation**: CCA with $q = 1$.
- [Supplement] **Discriminant analysis**: with $\mathbf{Y}$ a set of group indicator (dummy) variables, the canonical variates of $\mathbf{X}$ give the same directions as Fisher's discriminants (Ch. 10).
- **PCA**: variance maximization within one set (eigendecomposition of $\mathbf{\Sigma}$) versus correlation maximization between two sets (SVD of $\mathbf{\Sigma}_{11}^{-1/2}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1/2}$).

6 Selected exercises

Motivation. The exercises give a square-root-free way to compute canonical variates, two structures where the canonical correlation is found in closed form, and the likelihood ratio derivation of the test of $\mathbf{\Sigma}_{12} = \mathbf{0}$ quoted above.

Exercise 8.1 (Computing without square roots; J&W 10.4)

Let $\mathbf{M} = \mathbf{\Sigma}_{11}^{-1/2}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1/2}$ as in Theorem 8.2 and $\mathbf{K} = \mathbf{\Sigma}_{11}^{-1}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}$.
(a) Show that if $\mathbf{M}\mathbf{M}^\top\mathbf{e} = \lambda\mathbf{e}$ with $\mathbf{e} \neq \mathbf{0}$, then $\mathbf{K}(\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e}) = \lambda\,\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e}$, and that $\mathbf{K}$ and $\mathbf{M}\mathbf{M}^\top$ have the same characteristic polynomial.
(b) Deduce that $\mathbf{K}\mathbf{g}_k = \rho_k^2\mathbf{g}_k$ and that $\mathbf{h}_k = \rho_k^{-1}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}\mathbf{g}_k$ for $k \le r$.

Solution.

(a) $\mathbf{K} = \mathbf{\Sigma}_{11}^{-1/2}(\mathbf{M}\mathbf{M}^\top)\mathbf{\Sigma}_{11}^{1/2}$, because $\mathbf{M}\mathbf{M}^\top = \mathbf{\Sigma}_{11}^{-1/2}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}\mathbf{\Sigma}_{11}^{-1/2}$. Hence $\mathbf{K}\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e} = \mathbf{\Sigma}_{11}^{-1/2}\mathbf{M}\mathbf{M}^\top\mathbf{e} = \lambda\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e}$, and $\mathbf{\Sigma}_{11}^{-1/2}\mathbf{e} \neq \mathbf{0}$ since $\mathbf{\Sigma}_{11}^{-1/2}$ is invertible. Moreover

$$|\mathbf{K} - \lambda\mathbf{I}| = |\mathbf{\Sigma}_{11}^{-1/2}(\mathbf{M}\mathbf{M}^\top - \lambda\mathbf{I})\mathbf{\Sigma}_{11}^{1/2}| = |\mathbf{M}\mathbf{M}^\top - \lambda\mathbf{I}|,$$

so the eigenvalues of the non-symmetric $\mathbf{K}$ are real and equal to $\rho_1^2, \ldots, \rho_r^2$ and zeros.

(b)

Step 1. With $\mathbf{e} = \mathbf{a}_k$ and $\lambda = d_k^2 = \rho_k^2$ in (a), $\mathbf{K}\mathbf{g}_k = \rho_k^2 \mathbf{g}_k$ for $\mathbf{g}_k = \boldsymbol{\Sigma}_{11}^{-1/2}\mathbf{a}_k$.

Step 2. From $\mathbf{M} = \sum_j d_j \mathbf{a}_j \mathbf{b}_j^\top$ and $\mathbf{a}_j^\top \mathbf{a}_k = \delta_{jk}$, $\mathbf{M}^\top \mathbf{a}_k = d_k \mathbf{b}_k$. Hence for $k \le r$

$$\mathbf{h}_k = \boldsymbol{\Sigma}_{22}^{-1/2}\mathbf{b}_k = \frac{1}{\rho_k}\boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{22}^{-1/2}\boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1/2}\mathbf{a}_k = \frac{1}{\rho_k}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\mathbf{g}_k.$$

So one solves the eigenproblem of $\mathbf{K}$, normalizes $\mathbf{g}_k^\top \boldsymbol{\Sigma}_{11}\mathbf{g}_k = 1$, and obtains $\mathbf{h}_k$ by regression of the $\mathbf{Y}$-set on ξ_k. ■

Exercise 8.2 (Equicorrelation between and within sets; J&W 10.7)

Let $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$ be standardized with every correlation (within and between the sets) equal to $\rho \neq 0$:

$$\mathbf{R}_{11} = (1-\rho)\mathbf{I}_p + \rho \mathbf{1}_p \mathbf{1}_p^\top, \quad \mathbf{R}_{22} = (1-\rho)\mathbf{I}_q + \rho \mathbf{1}_q \mathbf{1}_q^\top, \quad \mathbf{R}_{12} = \rho \mathbf{1}_p \mathbf{1}_q^\top,$$

with the joint matrix positive definite. Find the canonical correlations and the first pair of canonical variates. Specialize to $p = q = 2$.

Solution.

Step 1 (action on 1). Let $\alpha_p = 1 + (p-1)\rho$. Then $\mathbf{R}_{11}\mathbf{1}_p = \alpha_p \mathbf{1}_p$ with $\alpha_p > 0$ (positive definiteness), so $\mathbf{1}_p$ is an eigenvector of $\mathbf{R}_{11}$ and, by Theorem 2.10, $\mathbf{R}_{11}^{-1/2}\mathbf{1}_p = \alpha_p^{-1/2}\mathbf{1}_p$. Likewise $\mathbf{R}_{22}^{-1/2}\mathbf{1}_q = \alpha_q^{-1/2}\mathbf{1}_q$.

Step 2 (the matrix M). Using symmetry of $\mathbf{R}_{22}^{-1/2}$,

$$\mathbf{M} = \mathbf{R}_{11}^{-1/2}\rho \mathbf{1}_p \mathbf{1}_q^\top \mathbf{R}_{22}^{-1/2} = \frac{\rho}{\sqrt{\alpha_p \alpha_q}} \mathbf{1}_p \mathbf{1}_q^\top = \frac{\rho\sqrt{pq}}{\sqrt{\alpha_p \alpha_q}}\, \mathbf{a}\mathbf{b}^\top, \qquad \mathbf{a} = \frac{\mathbf{1}_p}{\sqrt{p}},\ \mathbf{b} = \frac{\mathbf{1}_q}{\sqrt{q}}.$$

This is a rank-one SVD (replace $\mathbf{b}$ by $-\mathbf{b}$ if $\rho < 0$), so by Theorem 8.2

$$\rho_1 = |\rho| \sqrt{\frac{pq}{\{1+(p-1)\rho\}\{1+(q-1)\rho\}}}, \qquad \rho_2 = \cdots = 0.$$

Step 3 (variates). $\mathbf{g}_1 = \mathbf{R}_{11}^{-1/2}\mathbf{a} = \mathbf{1}_p/\sqrt{p\alpha_p}$, so $\xi_1 = \sum_i X_i/\sqrt{p\alpha_p}$, the standardized sum (indeed $\operatorname{Var}(\mathbf{1}_p^\top \mathbf{X}) = \mathbf{1}_p^\top \mathbf{R}_{11}\mathbf{1}_p = p\alpha_p$); similarly $\omega_1 = \operatorname{sign}(\rho)\sum_j Y_j/\sqrt{q\alpha_q}$. Check: $\operatorname{Corr}(\mathbf{1}^\top\mathbf{X}, \mathbf{1}^\top\mathbf{Y}) = \rho pq/\sqrt{p\alpha_p\, q\alpha_q}$, which is ρ_1 up to sign.

For $p = q = 2$: $\rho_1 = 2|\rho|/(1+\rho)$, $\xi_1 = (X_1 + X_2)/\sqrt{2(1+\rho)}$ and $\omega_1 = \operatorname{sign}(\rho)(Y_1 + Y_2)/\sqrt{2(1+\rho)}$. ■

Exercise 8.3 (Correlation for angular data; J&W 10.8)

An angle θ_2 is recorded as $\mathbf{Y} = (\cos\theta_2, \sin\theta_2)^\top$; assume $\operatorname{Cov}(\mathbf{Y}) \succ 0$.
(a) Show $\mathbf{b}^\top\mathbf{Y} = \sqrt{b_1^2 + b_2^2}\,\cos(\theta_2 - \beta)$, where $\cos\beta = b_1/\sqrt{b_1^2+b_2^2}$ and $\sin\beta = b_2/\sqrt{b_1^2+b_2^2}$.
(b) For a scalar X, show that the canonical correlation is $\rho_1 = \max_\beta \operatorname{Corr}\{X, \cos(\theta_2 - \beta)\}$.
(c) If also $\mathbf{X} = (\cos\theta_1, \sin\theta_1)^\top$, show $\rho_1 = \max_{\alpha,\beta} \operatorname{Corr}\{\cos(\theta_1 - \alpha), \cos(\theta_2 - \beta)\}$.

Solution.

(a) With $c = \sqrt{b_1^2 + b_2^2} > 0$, $\mathbf{b}^\top\mathbf{Y} = c(\cos\beta\cos\theta_2 + \sin\beta\sin\theta_2) = c\cos(\theta_2 - \beta)$.

(b)

Step 1. For scalar X and $a \neq 0$, $\mathrm{Corr}(aX, \mathbf{b}^\top \mathbf{Y}) = \mathrm{sign}(a)\mathrm{Corr}(X, \mathbf{b}^\top \mathbf{Y}) = \mathrm{Corr}(X, \mathrm{sign}(a)\mathbf{b}^\top \mathbf{Y})$, so $\rho_1 = \max_{\mathbf{b}\neq\mathbf{0}} \mathrm{Corr}(X, \mathbf{b}^\top \mathbf{Y})$.

Step 2. By (a) and scale invariance of correlation, $\mathrm{Corr}(X, \mathbf{b}^\top \mathbf{Y}) = \mathrm{Corr}\{X, \cos(\theta_2 - \beta)\}$; as $\mathbf{b}$ ranges over $\mathbb{R}^2 \setminus \{\mathbf{0}\}$, β ranges over $[0, 2\pi)$. Both variances are positive because $\mathrm{Cov}(\mathbf{Y}) \succ 0$.

Choosing the canonical variate therefore means choosing a new origin β for the angle.

(c) Apply (a) to both $\mathbf{a}^\top \mathbf{X}$ and $\mathbf{b}^\top \mathbf{Y}$ and use scale invariance; the maximum over $(\mathbf{a}, \mathbf{b})$ becomes a maximum over the two origins (α, β). ■

Exercise 8.4 (The likelihood ratio test of $\boldsymbol{\Sigma}_{12} = \mathbf{0}$)

For a sample of size $n > p + q$ from $N_{p+q}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, derive the likelihood ratio statistic for $H_0 : \boldsymbol{\Sigma}_{12} = \mathbf{0}$ and show $\Lambda^{2/n} = |\mathbf{S}|/(|\mathbf{S}_{11}||\mathbf{S}_{22}|) = \prod_k (1 - \hat{\rho}_k^2)$, with pq degrees of freedom.

Solution [Supplement].

Step 1 (unrestricted). By Theorem 4.7, $\max L = (2\pi)^{-n(p+q)/2} |\mathbf{S}_0|^{-n/2} e^{-n(p+q)/2}$.

Step 2 (under H_0). If $\boldsymbol{\Sigma}_{12} = \mathbf{0}$, then $|\boldsymbol{\Sigma}| = |\boldsymbol{\Sigma}_{11}||\boldsymbol{\Sigma}_{22}|$ and $\boldsymbol{\Sigma}^{-1} = \mathrm{diag}(\boldsymbol{\Sigma}_{11}^{-1}, \boldsymbol{\Sigma}_{22}^{-1})$, so the density of each observation factorizes and $L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = L_1(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11}) L_2(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$, the likelihoods of the $\mathbf{X}$- and $\mathbf{Y}$-samples. The two factors have separate parameters; Theorem 4.7 applied to each gives

$$\max_{H_0} L = (2\pi)^{-n(p+q)/2} |\mathbf{S}_{0,11}|^{-n/2} |\mathbf{S}_{0,22}|^{-n/2} e^{-n(p+q)/2}.$$

Step 3 (the ratio). $\Lambda^{2/n} = |\mathbf{S}_0|/(|\mathbf{S}_{0,11}||\mathbf{S}_{0,22}|)$, and the factors $(\frac{n-1}{n})^{p+q}$ cancel, so $\Lambda^{2/n} = |\mathbf{S}|/(|\mathbf{S}_{11}||\mathbf{S}_{22}|)$. The determinant identity above turns this into $\prod_k (1 - \hat{\rho}_k^2)$.

Step 4 (degrees of freedom). H_0 removes exactly the pq free entries of $\boldsymbol{\Sigma}_{12}$, so $-2 \log \Lambda \dot{\sim} \chi^2_{pq}$. For $q = 1$ the test is the test of zero multiple correlation; with blocks of size one, Exercise 7.1 is the analogous test for p blocks. ■